

# Seongmin Jung

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## RESEARCH INTEREST

Multimodal Robot Learning, Vision-Language(-Action) Models, 3D Computer Vision, Embodied AI

## EDUCATION

### Seoul National University

*M.S. in Artificial Intelligence (Advisor: Prof. Jongwoo Lim)*

GPA: 4.25/4.30

- Master's Thesis: *Panoramic Vision-Language Grounding for Generalizable 3D Scene Understanding*

Sep 2024 – Aug 2026

Seoul, South Korea

### Yonsei University

*B.S. in Electrical and Electronic Engineering*

GPA: 4.02/4.30 • Graduated with **High Honors (Top 3% in college)**

Mar 2019 – Feb 2024

Seoul, South Korea

### Case Western Reserve University

*Exchange Program*

Aug 2022 – May 2023

Cleveland, OH, USA

## HONORS AND AWARDS

### The National Scholarship for Science and Engineering

2019 – 2024

Korea Student Aid Foundation (KOSAF)

Seoul, South Korea

- Full merit-based scholarship awarded to top STEM students by the South Korean government, covering full tuition.

### High Honors

Fall 2019

Dept. of Electrical & Electronic Engineering, Yonsei University

Seoul, South Korea

### Highest Honors

Spring 2019

Dept. of Electrical & Electronic Engineering, Yonsei University

Seoul, South Korea

## PUBLICATIONS

- [1] **Seongmin Jung\***, Seongho Choi\*, Gunwoo Jeon, Minsu Cho, and Jongwoo Lim. PanoGrounder: Bridging 2D and 3D with Panoramic Scene Representations for VLM-based 3D Visual Grounding. *ECCV 2026*  
ArXiv: <https://arxiv.org/abs/2512.20907>

## RESEARCH EXPERIENCES

### AI4CE Lab, New York University

Internship (Advisor: Prof. Chen Feng)

Remote / New York City, NY, USA

*Tactile-Informed Imitation Learning*

Nov 2025 – Present

- Developing visual-tactile compositional diffusion policies that fuse low-rate visual inputs with high-rate piezoelectric tactile signals for contact-rich manipulation.
- Constructed a LeRobot-based training pipeline, reimplementing baselines for integration with the custom robot hardware.

### Robot Vision Lab, Seoul National University

Graduate Research

Seoul, South Korea

*Solving Rubik's Cube with Bimanual Robot Hands*

Sep 2025 - Present

- Collected VR teleoperation data on a Fourier GR-1 bimanual humanoid robot in Isaac Lab using a Meta Quest 3.
- Fine-tuned an open-source VLA (GR00T N1.5) for stable grasping and rotation of Rubik's Cube faces.

*PanoGrounder*

Jan 2025 – Mar 2026

- Proposed a framework for VLMs reasoning over 3D Mesh/3DGS via panoramic rendering, enabling spatial understanding.
- Fine-tuned a  $\sim 17B$  VLM on 8x A100 GPUs using DeepSpeed/LoRA, injecting DINOv2 features via custom adapters.
- Achieved SOTA (+4.7% on Nr3D), robust generalization (+17.4% on ARKitScenes), and efficient inference ( $\sim 3.3s/query$ ).

*Neural SLAM with Point Cloud Projection for Faster RGB-D Tracking*

Sep 2024 - Dec 2024

- Proposed a point-NeRF-based differentiable camera pose tracker, replacing iterative NeRF queries along camera rays.
- Replaced the tracking module of Point-SLAM, achieving over  $3\times$  faster tracking while preserving trajectory accuracy.

### Biological Cybernetics Lab, Yonsei University

Undergraduate Thesis (Advisor: Prof. DaeEun Kim)

Seoul, South Korea

*Autonomous Navigation for High-Speed Ground Robot*

Jul 2023 – Dec 2023

- Implemented low-latency obstacle avoidance and path planning in C++, using ROS and LiDAR odometry.
- Deployed on a Raspberry-Pi-powered RC car achieving high-speed navigation at max 5m/s.

## PROFESSIONAL EXPERIENCE

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### Auron Tomography Intelligence System

3D Vision Engineer

Oct 2025 – Present  
Remote / Pittsburgh, PA, USA

- Developing 3D volumetric segmentation networks (e.g., 3D U-Net) for processing noisy, high-resolution voxel data.
- Building data workflow, 5-fold cross-validation, and clinical evaluation.
- Collaborating with medical professionals to define clinical problems and enhance performance for real-world deployment.

### Mobiltech

Research Assistant

Aug 2023 – Jun 2024  
Seoul, South Korea

*Enhancing Pose Tracking in Urban Scenes*

- Developed a LiDAR odometry pipeline in C++, using per-frame height maps of surrounding structures, improving trajectory accuracy and robustness in dense urban scenes.

*Isaac Sim Environment for Heterogeneous Multi-Robot Systems*

- Implemented a heterogeneous multi-robot simulation with ground and aerial platforms in Isaac Sim, integrating per-robot control and sensor setups in ROS.

## TEACHING EXPERIENCES

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### Digital Computer Concept and Practice (F37.202), *Seoul National University*

Teaching Assistant

Spring 2025, Spring 2026  
Seoul, South Korea

- Led weekly C++ programming lab sessions and developed course materials and assignments.

## PATENTS

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- [1] Y. Lee, **S. Jung**, J. Jo, and Y. Kim. 2023. Apparatus for collecting cigarette butts with IoT. Korea Patent 1028688440000, September 30, 2025.

## CLUB ACTIVITIES

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### CWRUbotix, *Robotics Club at Case Western Reserve University*

MATE ROV World Championship 2023

Dec 2022 – Jun 2023  
Cleveland, OH, USA

- Constructed underwater robot simulation for mission environment using ROS and Gazebo Simulation.
- Designed and implemented an Arduino-based wireless communication module for mission execution.

### VERY, *Entrepreneurship Club at Yonsei University*

Team Lead

Mar 2021 – Feb 2022  
Seoul, South Korea

- Established online music ensemble platform based on self-recording for college bands.
- Led a 4-member team, coordinating operation, design, and development throughout the project.

## LEADERSHIP

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### Korean Mentor

Office of International Affairs, Yonsei University

Sep 2023 – Aug 2024  
Seoul, South Korea

- Mentored 10 incoming study-abroad students, aiding in cultural and academic adaptation in Korea.

### President of Student Council

Dept. of Electrical & Electrical Engineering, Yonsei University

Nov 2020 – Oct 2021  
Seoul, South Korea

- Led key events including freshmen orientation and homecoming, organizing 300+ students.

## TECHNICAL SKILLS

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**Programming Languages:** Python, C/C++, MATLAB, Verilog

**Libraries & Frameworks:** PyTorch, DeepSpeed, LeRobot, OpenCV, Open3D

**Robotics & Simulation:** ROS 1 & 2, Isaac Sim, Isaac Lab, VR Teleoperation (Meta Quest3 + XR Robotics)

**Languages:** Korean (native), English (fluent)

- TOEFL iBT: 108 (Reading: 30, Listening: 29, Speaking: 24, Writing: 25)